Diaspora-Adjusted Importation Risk: Methods, Logic, and Interpretation

Technical note — FIFA World Cup 2026 importation risk analysis

1. What the core model gives us — and what it misses

The core Poisson model produces, for each source country c, venue city v, and disease d:

$$\lambda_{c,v,d} = N_{c,v} \cdot \rho_d \cdot \frac{C_{c,d}}{P_c} \cdot p_d$$

where $N_{c,v}$ is the projected number of travelers from c to v, ρ_d is the under-reporting correction, $C_{c,d}/P_c$ is reported per-capita incidence, and p_d is the probability of traveling while infectious. Summing across source countries gives the city-level importation intensity $\Lambda_{v,d} = \sum_c \lambda_{c,v,d}$, and the probability of at least one importation follows from the Poisson CDF:

$$P(\geq 1)_{v,d} = 1 - e^{-\Lambda_{v,d}}$$

This answers the question: *how many cases will arrive, and how likely is at least one?*

What it does **not** answer is: *what social environment do arriving cases enter?* An infectious traveler from Brazil who lands in Boston enters a city where 7.9% of all foreign-born residents are Brazilian — a large, socially connected community of co-nationals who share language, neighborhoods, community events, and World Cup watch gatherings. The same traveler landing in Kansas City enters a city with almost no such community. The Poisson intensity λ is identical in both cases if the travel volumes are the same; the downstream transmission potential is not.

The diaspora extension quantifies this difference.

2. Diaspora data and notation

We use the US Census Bureau American Community Survey (ACS) 5-year estimates (2019–2023), Table B05006 (*Place of Birth for the Foreign-Born Population*), for the Core-Based Statistical Area (CBSA) corresponding to each of the 11 WC host metro areas.

- $D_{c,v}$ — number of US residents born in country c currently living in metro area v (ACS estimate).

- FB_v — total foreign-born population of metro v (the row total of Table B05006). This is the denominator: it represents *all* immigrants in that city, regardless of origin country.
- $\kappa_{c,v} = \dfrac{D_{c,v}}{\mathrm{FB}_v}$ — **diaspora concentration**: the share of metro v's foreign-born population that was born in country c. This is a number between 0 and 1.

Concrete examples. Brazilians constitute 7.9% of Boston's foreign-born population ($\kappa_{\text{Brazil, Boston}} = 0.079$). Haitians constitute 8.8% of Miami's ($\kappa_{\text{Haiti, Miami}} = 0.088$). Ecuadorians constitute 4.8% of New York's ($\kappa_{\text{Ecuador, NY}} = 0.048$). For most country–city pairs κ is much smaller, often below 1%.

A critical point: κ is normalized by the *total foreign-born* population, not by the total metro population. This makes κ a measure of how concentrated the source-country community is *within the immigrant population* of that city — the relevant social network for co-national mixing.

3. Mechanism A: local social mixing at the venue city

3.1 Reasoning

When an infectious traveler from country c arrives at venue city v, they do not mix at random with the entire population. They are more likely to interact with co-nationals — through shared language, cultural spaces, restaurants, community events, and World Cup gatherings organized within the diaspora. We model this selective mixing by assuming that, upon arrival, each traveler's social contacts are drawn proportionally from the foreign-born population of the city.

3.2 The mixing assumption

Assumption: each infectious traveler from country c who arrives at city v interacts with the foreign-born population of v. Under proportional mixing, the probability that a randomly chosen interaction partner was born in country c (i.e., is a co-national) equals $\kappa_{c,v}$.

This is a simplification — actual mixing rates depend on neighborhood clustering, social network structure, and event attendance — but it provides a transparent, data-grounded lower bound on co-national contact probability.

3.3 Mathematics: Poisson thinning

Let $X_{c,v,d}$ be the number of infectious travelers from country c arriving at city v. From the core model:

$$X_{c,v,d} \sim \text{Poisson}(\lambda_{c,v,d})$$

Each of these $X_{c,v,d}$ travelers independently enters the co-national diaspora network with probability $\kappa_{c,v}$ (the mixing assumption above). Let $Y_{c,v,d}$ be the number of those travelers who actually enter the co-national network. By the **Poisson thinning theorem**:

$$Y_{c,v,d} \sim \text{Poisson}(\lambda_{c,v,d} \cdot \kappa_{c,v})$$

We define:

$$\Omega^A_{c,v,d} = \lambda_{c,v,d} \cdot \kappa_{c,v} = \lambda_{c,v,d} \cdot \frac{D_{c,v}}{\text{FB}_v} \tag{1}$$

$\Omega^A_{c,v,d}$ is the expected number of imported cases of disease d, from country c, that enter the co-national social network in city v. It is a valid Poisson rate for a *sub-population* of the original $\lambda_{c,v,d}$ arrivals.

The city-level aggregate across all source countries is:

$$\Omega^A_{v,d} = \sum_c \Omega^A_{c,v,d} = \sum_c \lambda_{c,v,d} \cdot \kappa_{c,v}$$

Because this is a sum of independent Poisson random variables, $\Omega^A_{v,d}$ is itself a Poisson rate, and the probability that at least one imported case enters *any* co-national network in city v is:

$$P^A(\geq 1)_{v,d} = 1 - e^{-\Omega^A_{v,d}} \tag{2}$$

3.4 How to read and compare the two probabilities

Figure 1 shows $P(\geq 1)_{v,d}$ — the probability that at least one case arrives at city v. Figure 4 shows $P^A(\geq 1)_{v,d}$ — the probability that at least one of those cases enters a co-national network. Both are derived from the same Poisson logic. Their difference is interpretable:

Example. Dengue at Miami: $P(\geq 1) \approx 1$ (virtually certain that at least one dengue case arrives). Suppose $P^A(\geq 1) = 0.60$. This means: *it is virtually certain that cases arrive, but there is only a 60% probability that at least one of those cases enters a co-national social network*. In 40% of scenarios, cases arrive but disperse into the general population — still a public health concern, but without the added amplification of concentrated co-national social ties.

The gap between $P(\geq 1)$ and $P^A(\geq 1)$ is the fraction of risk that is *diffuse* (general population) rather than *concentrated* (co-national network). Cities where this gap is small have highly concentrated diaspora communities relative to total importation intensity; cities where the gap is large have high importation volumes but dispersed or thin co-national networks.

Why are P^A values always lower than P? Because $\Omega^A_{v,d} \leq \Lambda_{v,d}$ always: $\kappa_{c,v} \leq 1$ for all c, v, so multiplying $\lambda_{c,v,d}$ by $\kappa_{c,v}$ can only reduce it. The diaspora network is always a sub-population of the full arrival stream; it cannot have more cases than the whole.

4. Mechanism B: diaspora return seeding

4.1 Reasoning

Mechanism A asks what happens when international travelers arrive at venue cities where co-nationals already live. Mechanism B runs in the opposite direction: US-resident diaspora members — already living in the US — travel *domestically* to a WC venue city to watch their country's matches, are exposed to infectious travelers from their home country at the venue, and then return to their home city. The risk lands not at the venue, but in the diaspora hub city.

This mechanism is entirely invisible to the core model, which only tracks international arrivals. A city that hosts no WC matches and receives few international visitors from a given country can still face secondary seeding through returning diaspora members.

4.2 Notation

Let v_{match} be the venue city where country c plays, and v_{home} be a hub city where a diaspora community from c lives ($v_{\text{home}} \neq v_{\text{match}}$).

4.3 Mathematics

The exposure source at v_{match} is $\lambda_{c,\,v_{\text{match}},\,d}$ — the expected number of infectious travelers from country c arriving at the match venue. This is the same Poisson rate computed by the schedule-driven model (M3). Diaspora members from v_{home} who attend the match are exposed to this infectious pool. Upon return to v_{home}, they may seed transmission among co-nationals there. The secondary seeding potential is:

$$\Omega^B_{c,\,v_{\text{match}},\,v_{\text{home}},\,d} = \lambda_{c,\,v_{\text{match}},\,d} \cdot \kappa_{c,\,v_{\text{home}}} = \lambda_{c,\,v_{\text{match}},\,d} \cdot \frac{D_{c,\,v_{\text{home}}}}{\text{FB}_{v_{\text{home}}}} \tag{3}$$

Important: Ω^B has the same algebraic form as Ω^A but a different interpretation. In Ω^A, both λ and κ refer to the *same city* (v); the question is how many of the arrivals at v contact co-nationals in v. In Ω^B, λ refers to the *match venue* (v_{match}) and κ refers to the *hub city* (v_{home}); these are different places. Ω^B is therefore better interpreted as a *relative seeding index* — it ranks hub cities by their exposure potential, proportional to both the infectious pressure at the venue and the density of the diaspora in the hub — rather than a strict Poisson rate for secondary cases.

The hub-city aggregate across all source countries and match venues is:

$$\Omega^B_{v_{\text{home}},\,d} = \sum_{c} \sum_{v_{\text{match}} \neq v_{\text{home}}} \Omega^B_{c,\,v_{\text{match}},\,v_{\text{home}},\,d}$$

5. Summary: what each quantity means

Quantity	Definition	Plain interpretation
$\lambda_{c,v,d}$	$N_{c,v} \cdot \rho_d \cdot (C_{c,d}/P_c) \cdot p_d$	Expected infectious travelers from c arriving at v.
$P(\geq 1)_{v,d}$	$1 - e^{-\Lambda_{v,d}}$	Probability ≥ 1 case arrives at city v.
$\kappa_{c,v}$	$D_{c,v}/\mathrm{FB}_v$	Share of city v's immigrants born in c.
$\Omega^A_{c,v,d}$	$\lambda_{c,v,d} \cdot \kappa_{c,v}$	Expected cases *entering co-national network* in v. Valid Poisson rate (thinned from λ).
$P^A(\geq 1)_{v,d}$	$1 - e^{-\Omega^A_{v,d}}$	Probability $\geq$ 1 case enters a co-national network in v. **Directly comparable to $P(\geq 1)_{v,d}$.**
$\Omega^B_{c,v_m,v_h,d}$	$\lambda_{c,v_m,d} \cdot \kappa_{c,v_h}$	Relative seeding index for hub city v_h from match at v_m. Ranks cities; not a probability.

Key constraint. $\Omega^A_{c,v,d} \leq \lambda_{c,v,d}$ always, because $\kappa_{c,v} \leq 1$. The diaspora framework does not create additional cases — it partitions the existing importation risk into the fraction that enters co-national networks (Ω^A) versus the fraction that disperses into the general population ($\lambda - \Omega^A$). The sum $\Omega^A + (\lambda - \Omega^A) = \lambda$ is exact.